

Supplementary Note S1 — Provenance, balancing derivation, and analysis self-checks

Companion to `paper/manuscript.md`. Every number in the manuscript is regenerated from one

file, `flymind.brain` (FLYMIND4), by the commands listed here. This note is the reproducibility

half of the paper; `results/PROVENANCE.md` holds the full-length version including file hashes.

S1.1 Source data

dataset	released file	local file	bytes	
MaleCNS v1.0 flat connectome	connectome- weights-male-cns- v1.0-minconf- 0.5.feather	malecns/connectome- weights.feather	1,051,241,946	E35DA783D1C686B2B58B3F
MaleCNS v1.0 annotations	body-annotations- male-cns-v1.0- minconf- 0.5.feather	malecns/body- annotations.feather	14,483,314	2177E246113E4CFBF1E77
MaleCNS v1.0 transmitter predictions	body- neurotransmitters- male-cns- v1.0.feather	malecns/body- neurotransmitters.feather	43,282,834	95C9289220663ABEB3409

Release bucket `gs://flyem-male-cns`; download page `https://male-cns.janelia.org/download`.

Derived intermediates: `malecns/edges.bin` (3,037,133,688 bytes; int64 count + 151,856,684 rows

of 20 bytes, SHA256 59E43C67...17BD0D), `malecns/annotations.tsv` (211,577 rows),

`malecns/transmitters.tsv` (164,401 rows). Feather/Parquet were read with purpose-written pure

Python readers (`tools/feather_lite.py`, `tools/parquet_lite.py`); no third-party dependency is

used anywhere in the pipeline.

S1.2 Pipeline

'''

export_edges.py / export_annotations.py / export_transmitters.py (Python 3.14.7)

↓ edges.bin, annotations.tsv, transmitters.tsv

CircuitBuilder (C#, .NET 8.0.404) → flymind.brain (FLYMIND4, 21,383,125 bytes,
SHA256 FC6B544C...836E0C8A, ≈140 s)

↓

CircuitPreview --exp / --dose / --pp / --asym / --propagate → per-trial CSV + stdout

↓

compare_trials.py → descriptive stats, pairwise tests, equivalence + Bayes factors

'''

Exact commands for every table are in `results/raw/00-commands.txt`.

S1.3 Circuit blob (FLYMIND4)

'''

magic 8s "FLYMIND4"

uint32 n_neurons, n_synapses, 0

uint32 pop_counts[12]

uint32 seed_count[6] sugar, motor, odor, selfmotion, centralcomplex, visualrelay

per neuron 24 B uint64 body id | float32 soma x,y,z (μm) | uint8 population | uint8 cx
family | 2 pad

CSR uint32 offsets[n+1] ; uint32 targets[n_syn] ; float32 weights[n_syn] (signed)

plasticity uint8 kc_mbon_mask[n_syn] ; float32 base_weight[n_syn]

seeds 6 × uint32 index arrays

metadata uint32 json_len + UTF-8 JSON

'''

Integrity checks executed on load: `offsets[0] = 0, offsets[-1] = n_synapses`
(1,615,753),

offsets monotone non-decreasing, all targets inside $[0, n)$. Native weight distribution

($|w|$ before κ): min 0.000275, p25 0.000275, median 0.000550, p75 0.001650, max 0.530475,

mean 0.001887 mV; 540,052 of 1,615,753 synapses negative (33.42%); 3,206
KC→MBON synapses

flagged plastic.

S1.4 Eq. S1 — per-step drive balancing

Let the visual pool have n cells, split into the body patch B ($|B| = pn$) and the surround

S . The natural condition delivers level $d_i^0 = f(t)$ to every cell, where $f(t)$ is the optic-flow

or rectified-yaw drive. The manipulation replaces the patch's drive by $\rho \cdot g(t)$, where $g(t)$ is the

patch's own signal (identical to $f(t)$ for `self`, a block-shuffled copy for `decorr`, or the

blend $(1-\alpha)f + \alpha \cdot g$ for the conflict-dose arms). The surround level s is solved from the levels

actually delivered in that step:

$$\sum_i d_i = \sum_{\{B\}} \rho \cdot g + s \cdot \sum_{\{S\}} f = \sum_{\{all\}} f \Rightarrow s = (\sum_{\{all\}} f - \sum_{\{B\}} \rho \cdot g) / \sum_{\{S\}} f$$

and the patch is capped so that it can never absorb more than the whole budget,

$\rho \cdot g \leftarrow \rho \cdot g \cdot \min(1, \sum_{\{all\}} f / \sum_{\{B\}} \rho \cdot g)$, which is exact because the rectifier is linear in a

non-negative factor. Injection is then Bernoulli per cell with $p_i = 150 \text{ Hz} \cdot \Delta t \cdot d_i$ (p clamped at 1,

levels are not; $p_{\text{max}} = 0.075$ so levels up to ≈ 13 are representable). Three consequences, all

verified in the matching tables: the *expected* event total is identical in every arm, the

realised totals differ only by Bernoulli noise, and no arm can deliver more than the natural

condition. The stimulus waveform, its shuffled copy, the odour background and the sampled CX

pairs are drawn from one shared stream per trial index (`Random(90000 + trial)`); arm-specific

streams are used only for the pool draws.

S1.5 Bayes-factor self-check

`python tools/compare_trials.py --selftest` prints the JZS two-sample Bayes factor

(Cauchy prior scale $r = 0.707$) against the analytic anchor at $t = 0$:

'''

$t = 0.0$ BF10 = 0.3973 BF01 = 2.5171 (analytic: $1/E[(1+n_{\text{eff}}g)^{-1/2}] = 1/0.3974$, $n_{\text{eff}} = 5$)

$t = 1.0$ BF10 = 0.5634 BF01 = 1.7748

$t = 2.0$ BF10 = 1.4964 BF01 = 0.6683

$t = 4.0$ BF10 = 34.4220 BF01 = 0.0291

'''

Design ceiling (BF01 at $t = 0$ against trials per arm): $n = 10 \rightarrow 2.52$; $16 \rightarrow 2.97$; $20 \rightarrow 3.24$;

$30 \rightarrow 3.81$; $50 \rightarrow 4.74$. This is the maximum evidence for the null that this design can produce,

and it is why the manuscript reports the Bayes factor together with the equivalence tests rather

than either alone.

S1.6 File index for the reported numbers

manuscript element	file
Fig. 1A–1C, circuit census / calibration / pathways	circuit-schematic.md, results/raw/calibration-kappa-sweep.txt
Fig. 1D–1F, manipulation and balancing	results/raw/A3-flow-matching.csv, B3-heading-matching.csv
Fig. 2, propagation	results/raw/propagation.stdout.txt
Fig. 3, optic flow	results/raw/A3-flow-

	{compare, pairs, trials, summary}.csv
Fig. 4, yaw null + conflict dose	results/raw/B3-heading-, <i>E-conflict</i> -, F-n20-*
Fig. 5A, flat rate and drive	results/raw/E-conflict-{trials, summary}.csv
Fig. 5B, gain replication	results/raw/G-k150-, <i>H-k600</i> -
Fig. 5C, Bayes factors at n = 20	results/raw/F-n20-equivalence.csv
Fig. S1, patch-size dose and positive controls	results/raw/C3-dose-dose-, <i>D-controls</i> -
Fig. S2, equivalence bounds	results/raw/F-n20-equivalence.csv
Fig. S3, Bayes-factor design ceiling	tools/make_main_figures.py (supp3), analytic anchor in §S1.5
Fig. S4, the drive-matching trap	results/raw/B-heading-trials.csv (unbalanced) vs B3-heading-trials.csv
Table 1 (n = 10 equivalence)	results/raw/B3-heading-equivalence.csv
Table 2 (n = 20 Bayes factors)	results/raw/F-n20-equivalence.csv
Table 3 (conflict dose)	results/raw/E-conflict-trials.csv analysed by tools/conflict_trend.py
table S4 (gain replication)	G-k150-trials.csv, H-k600-trials.csv likewise

S1.7 Conflict-dose trend analysis

`python tools/conflict_trend.py results/raw/E-conflict-trials.csv` maps arm id $\rightarrow \alpha$

(self 0, conf25 0.25, conf50 0.5, conf75 0.75, decorr 1), ranks the trial-level values

of every metric, and reports Spearman ρ against α with a permutation P (4,000 relabellings,

seed 12345). The same command run on G-k150-trials.csv and H-k600-trials.csv gives the two

extra gains. Metrics whose ρ is not reported had fewer than eight finite trial values (the two

legacy anatomical-gain readouts, which do not converge on every trial).